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BrainAGE predicts Brain Reserve estimated using a residual approach in Alzheimer's continuum

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Introduction

Resilience refers to multiple reserve-related processes, e.g. cognitive reserve(CR) and brain reserve (BR) (Stern et al. 2020; Ossenkoppele et al. 2020), while facing alterations in the brain, i.e. aging or neurodegenerative disease such as Alzheimer's disease (AD) **(Figure 1)**. The Brain Age Gap Estimation (BrainAGE) algorithm is developed to estimate morphological brain age, which was suggested to be affected by various neurological and psychiatric diseases, including preclinical and clinical AD (Gaser et al. 2013). However, the associations of CR and BR with BrainAGE remain unclear.

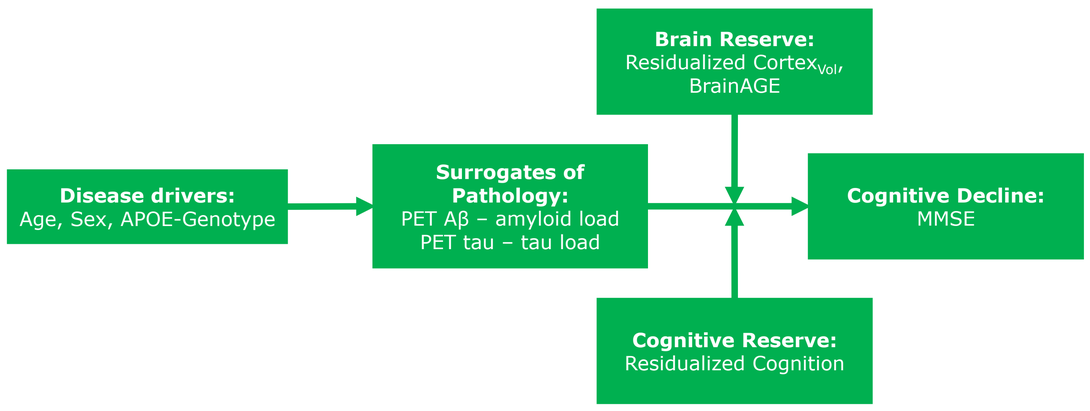

Figure 1. Theoretical model integrating CR and BR in estimating longitudinal cognitive trajectories.

Results

BrainAGE predicted higher residual BR marker (BRM; β=0.13, p=0.002, **Figure 2**), however, it was not associated with residual CR marker (CRM; β=0.08, p=0.08). Linear mixed models **(Figure 3)** revealed that higher CRM was associated with a slower cognitive decline over time (p=0.002, p=0.003; for models with BRM and BrainAGE, respectively). Significant three-way interactions were revealed between CR and BR over time on cognition (BRM*CRM*Time p=0.008; BrainAGE*CRM*Time, p=0.015), after models were adjusted for age, sex, years of education, A/T groups and study sites.

Methods

In this study, we included N=195 (Amyloid beta) Aβ and tau PET negative andcognitively normal healthy controls (71 years, 119 female) and N=264 Aβ PET-positive individuals (74 years, 142 female) as Alzheimer's continuum from Alzheimer's Disease Neuroimaging Initiative (ADNI) study. The participants underwent structural T1-MPRAGE MRI imaging, Aβ PET and tau PET. BR was calculated as the residualized global cortical volume by tau PET uptake. In contrast, CR was defined as the residualized global cognitive composite score by Tau and Aβ PET uptakes as predictors.

BrainAGE was estimated as the gap between the brain age estimated from MR images and the actual chronological age (Franke et al. 2010). As cognitive assessment we included data of mini-mental-state-examination (MMSE) at follow-up visits after one-year (N=170) and two-years (N=68) from the baseline imaging.
The associations between BrainAGE and BRM or CRM were tested in separately conducted multilinear regression analyses, while the interaction terms BRM × CRM × time points or BrainAGE x CRM x time points on MMSE were tested using separate linear mixed models.

Conclusion

BrainAGE estimated a residual marker of BR and successfully replaced it in longitudinal analysis, moderating the relationship between CR, time and cognition. Our results highlight the clinical value of BrainAGE as a proxy of brain reserve, having a significant advantage, being a unimodal approach compared to the residual approach to estimate BR.

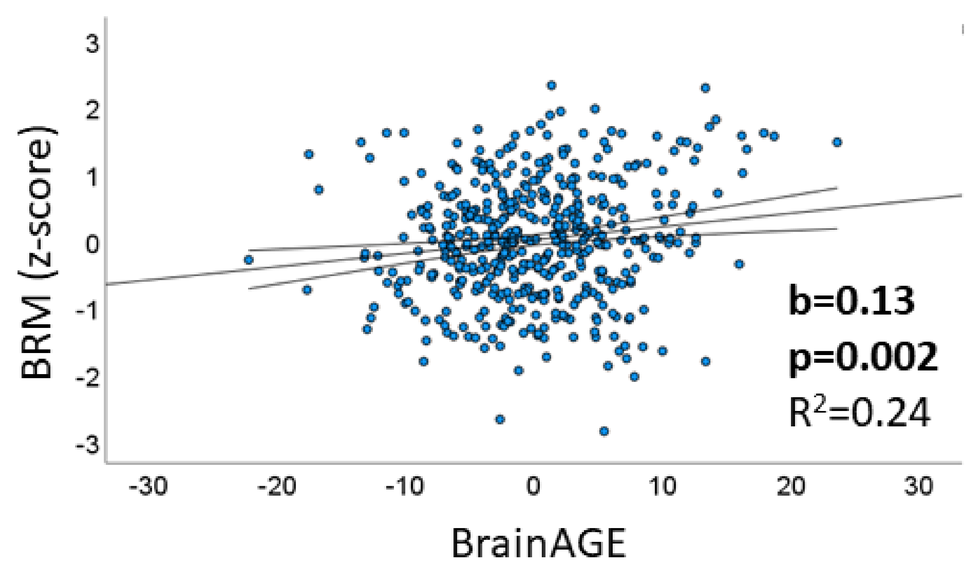

Figure 2. Association between BRM and BrainAGE. Lines: 95% confidence intervals..

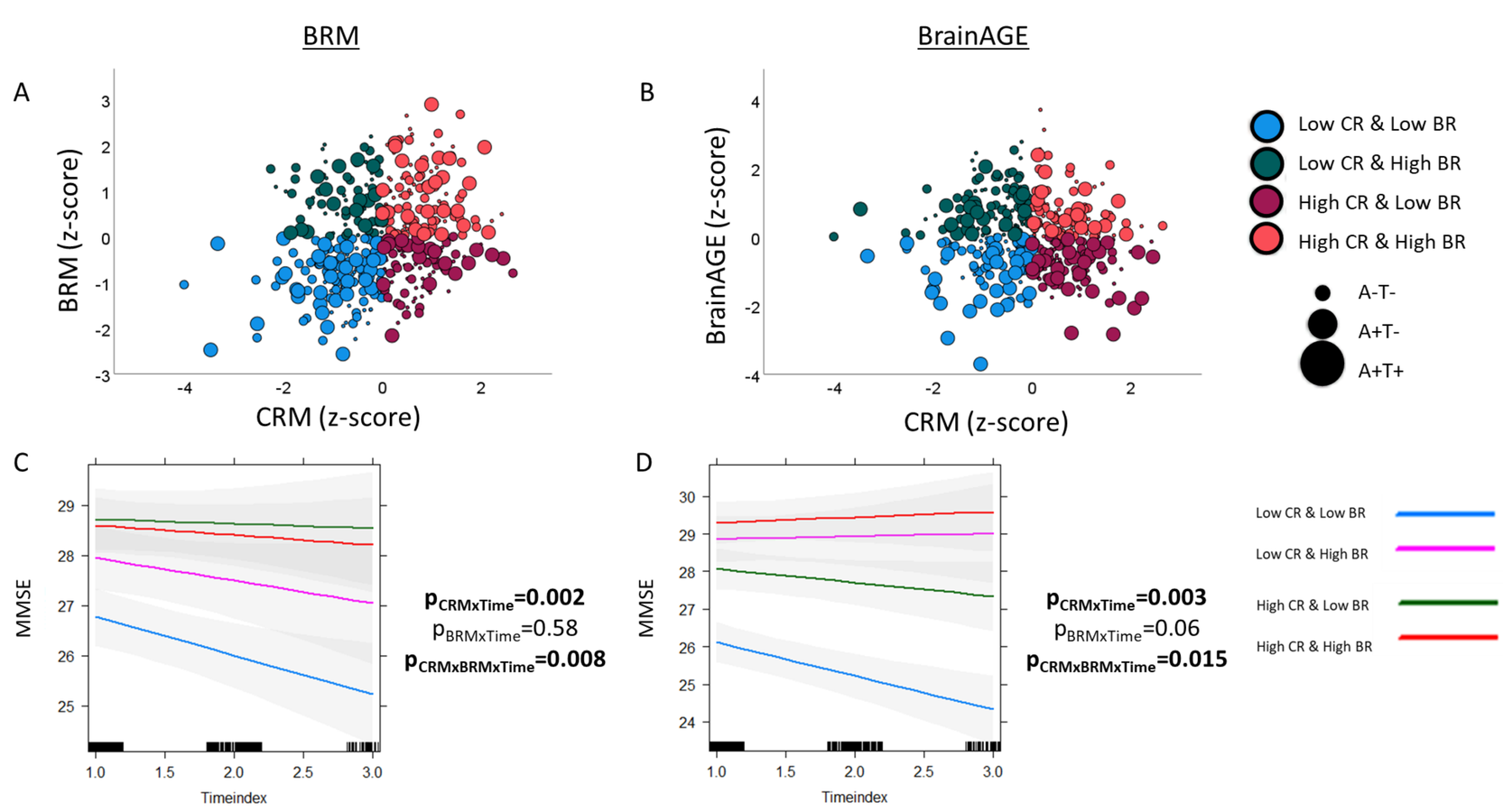

Figure 3. The scatter plots show binarized BR and CR groups via median split, considering BRM (A) and BrainAGE (B) as proxies of BR. Plots showing the longitudinal change in BR and CR groups, using BRM (C) and BrainAGE (D) as proxy of brain reserve.

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ADNI Alzheimer's Disease Neuroimaging Initiative